

Project Process Book

Datasets

<https://www.kaggle.com/datasets/thedevastator/spotify-tracks-genre-dataset>

<https://www.kaggle.com/yamaerenay/spotify-dataset-19212020-600k-tracks>

Overview and Motivation

Over the past century, music has transformed dramatically in response to technological innovation, cultural movements, and evolving listener preferences. Modern streaming platforms now capture both large-scale listening behaviors and detailed audio analyses, making it possible to quantitatively study what makes certain songs popular. This project aims to investigate how the musical characteristics of popular tracks have changed from 1921 to 2020 by analyzing a comprehensive Spotify dataset containing over 446,000 songs. We focus on two major categories of musical attributes: technical features, such as tempo, loudness, duration, and key. And perceptive features, Spotify's machine-generated measures of qualities like danceability, energy, and valence. By examining how these characteristics correlate with popularity and how their distributions shift over time and across genres, our goal is to understand which attributes define popular music in different eras.

Related Work

Our inspiration for this project grew out of our own music tastes and the annual excitement around Spotify Wrapped. Each year, Spotify Wrapped summarizes listeners' habits by highlighting how their preferences shift across genres, moods, and styles throughout the year. It's a feature people look forward to and often share on social media, because it offers a personalized snapshot of their musical identity. We wanted to extend this idea beyond the individual level. Rather than examining how one person's preferences change over a single year, we set out to explore how society's musical tastes have evolved over the past century. By scaling this kind of analysis up to decades of music and hundreds of thousands of tracks, our project aims to uncover long-term trends in what listeners find popular and how the defining characteristics of music have changed over time.

Questions

We began with the overarching question; "How have the musical characteristics of popular tracks changed over the past century?" We split this into two separate questions; "What are the (technical and perceptive) features of popular music" and "How have these features changed over time." Naturally these two questions beget the three graphs we used.

We found that it was not feasible nor necessary to include all the features of our dataset, so we asked, "what features are significant." This filter took away the features we were most interested in and left the more trivial ones (e.g. whether or not the track is acoustic, live, etc.). So, we shifted to "what features are audiophiles most interested in."

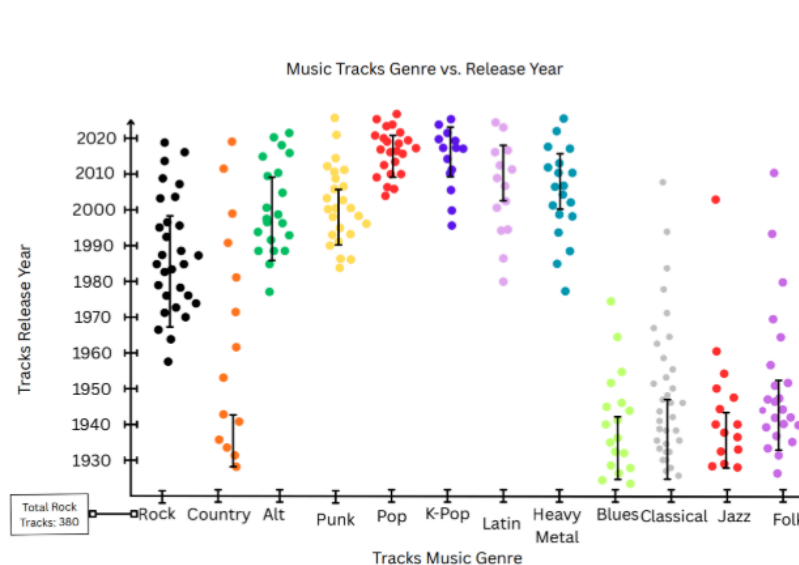
To expand our analysis, we are also prepared to ask, “how will our patterns shift when we consider genre.”

Data

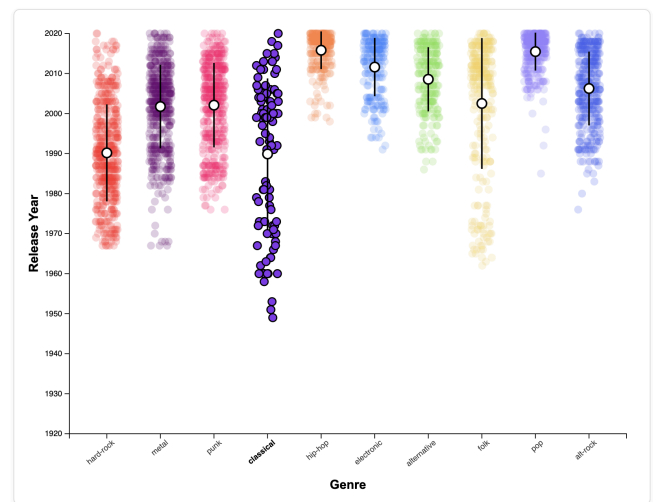
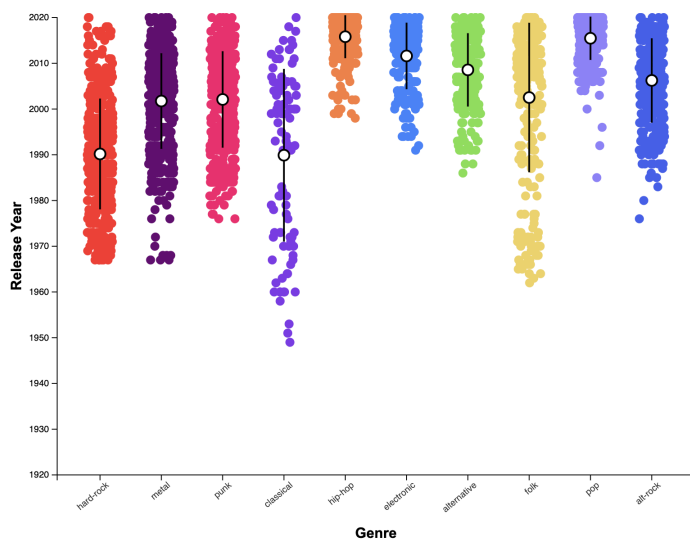
Our analysis is based on the Spotify Tracks Dataset (1921–2020) collected by Yamac Eren Ay using the Spotify Web API and publicly available on Kaggle. The dataset contains approximately 446,474 tracks from over 114,000 artists, along with both technical audio features and perceptive attributes generated by Spotify’s machine-learning models. Because the data was gathered through the API rather than scraped manually, it arrives in a structured format with consistent attribute definitions. However, several cleanup steps were necessary before analysis. Our main dataset contained around 500,000 tracks. However, it did not include the attributes for the genre of music. So we had to combine it with another dataset which only contained around 100,000 of the tracks. We combined the two datasets using track_id and cleaned the data removing any null values for genre. In our graphs we analyzed different tracks by their release year so we had to extract the year from release_date.

Design Evolution

For our initial design of our first graph we wanted to have one major graph that showed how different genre popularity changed over the decades. So our initial prototype was this:



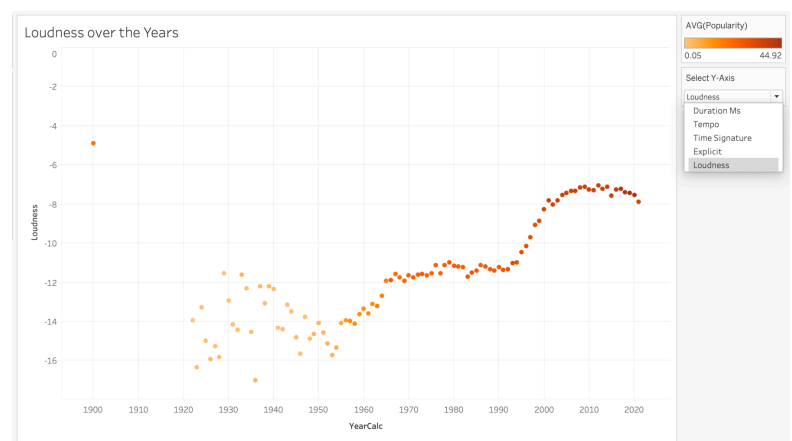
However, as we started to clean the data and work through the rest of our graphs we realized it would need to be altered. While this graph does show some insights into the question of how music taste changed over time, it could be better. We had needed the graphs to be interactive so we decided to make the genre a selective option on our first graph, and then the others reflect that. For the first graph, the design is nearly identical to the one shown above. However, after selecting a genre it would then focus on that one. It would still be a strip plot, but instead the release year would be on the x-axis, and each point an individual track.

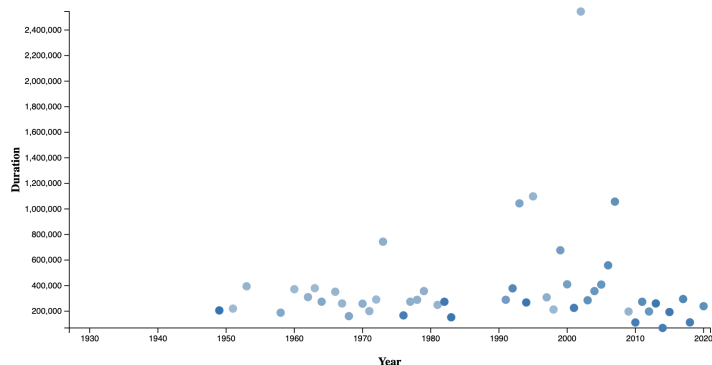


This was the final result of our genre graph. The image on the left shows the graph as it loads, with no filters active. It matches our original idea of the graph showing every point, by genre, with the median release year marked. In the image on the right, our interactivity for this graph is shown. When one genre is selected, the points associated with it are highlighted. This allowed for more interactivity throughout our dashboard to be changed, which meant more updates for the plans for our other two graphs as well.

For the next two visualizations, we focused on the questions “What are the technical and perceptive features of popular music?” Naturally this question begets the next two graphs we used; one that plots technical features and one that plots perceptive features. We found that it was not feasible nor necessary to include all the features of our dataset, so we asked, “what features are significant.” This filter took away the features we were most interested in and left the more trivial ones (e.g. whether or not the track is acoustic, live, etc.). So, we shifted to “what features are audiophiles most interested in.”

The second visualization analyzes the technical features against popularity. These features are explicitness, duration, tempo, time signature, and loudness. Throughout the process the design remained similar to what we initially planned it to be. Our initial design idea was to have a dropdown menu that allowed you to filter the graph based off of the features and see how that affected the popularity of music over the years. With the feature on the y-axis and popularity shown through a variation in color, we were able to have a good analysis of the question.





With the new implementation of filtering by genre, we edited this design slightly to interact well with the main graph. Now, the user can view the graph based on genre and feature (explicitness, duration, tempo, time signature, and loudness).

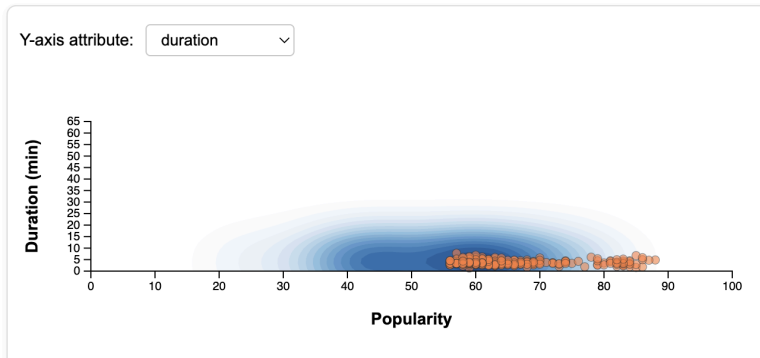
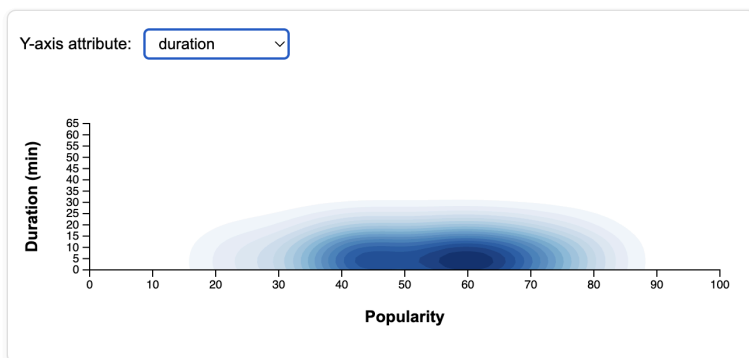
You aren't able to see our dropdown menu in this image, but it is as shown at the bottom of the screen. In the

future this will probably be moved around for a more user friendly experience.

Choose a genre: Select Y-axis attribute:

Further updates for this graph came along the way. Once the genre filter was fully implemented, we realized that since we were already seeing the songs by year, it might be better to see the technical attributes versus popularity. This change to the x-axis to show popularity instead of time, changed the overall interaction some. It led to more interaction updates, such as when a point was clicked on the genre graph, it would be highlighted on the technical features graph, and vice versa. This allows for the user to learn a lot about one song: its genre, what year it came out, and its popularity in respect to the y-axis attribute that the user has selected (ex: duration). These interactions increased the overall complexity of the graph and the dashboard overall, and allowed for the user to further dive into the question: "How have the musical characteristics of popular tracks changed over the past century?"

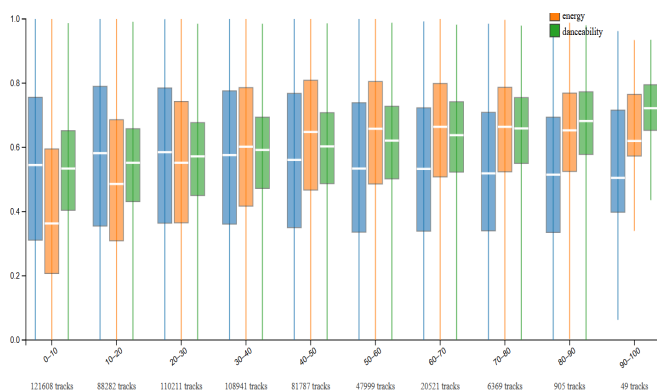
After these changes were made, the dashboard showed two graphs, and the placeholder for the genre graph that read "← Please choose a genre from the left graph". This didn't look great, so we decided that adding a contour graph to start would be better. Adding this might also help the user in comparing trends of a genre to the overall data that includes all the songs. We decided to keep it as an underlying graph so that when a genre was selected, that genre would be shown over top of the contour graph. Once that was added it really started coming together, and left us with our final product for the technical features graph.



Another thing that acted as an addition to our dashboard that came from the x-axis change, was the time slider. Once the x-axis on the technical graph was changed to Popularity instead of Year (time), we decided that integrating a time slider might be helpful to the user to see how an attribute might have changed over time. Although the user can see some of this information by selecting a single point and checking the genre graph, or hovering over it, to see when it was released, the time slider allows for a greater scale of comparison and exploration for the user.

The third visualization plots popularity against the three non-binary perceptive features: valence, danceability, and energy. These are the three most interesting perceptual features and the only perceptual features that vary naturally across all songs. To allow the user to view the interplay between these three features, we chose to display them together on the graph.

Only having three features and these features being normalized allowed them to share the same y-axis and be presented together on the graph. Staying consistent with the plot above, the x-axis was chosen to display the popularity. Unlike the above plot, this graph uses buckets and averages because a scatterplot was not visually clear. We chose to use a mixed bar chart because a scatter plot would make it more difficult to see trends as the only separation between the variables would be color. Grouping the data into buckets and using averages removes much of the individual information but increases the clarity of the overall trend. Still using color to separate the features continues to increase visibility and visual isolation.

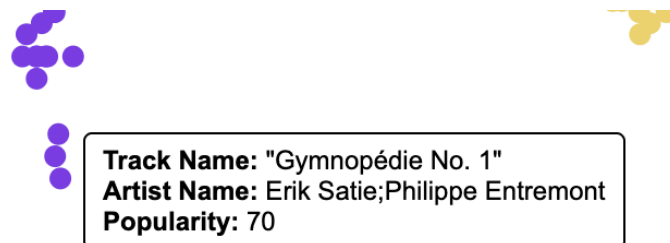


An alternative design for the third visualization (the perceptive features) was to replace the bars with boxplots. This would give the viewer substantially more information – a full spread for each bucket rather than only the average. This design, however, was too visually cluttered and added details that did not significantly contribute to the whole of the dashboard.

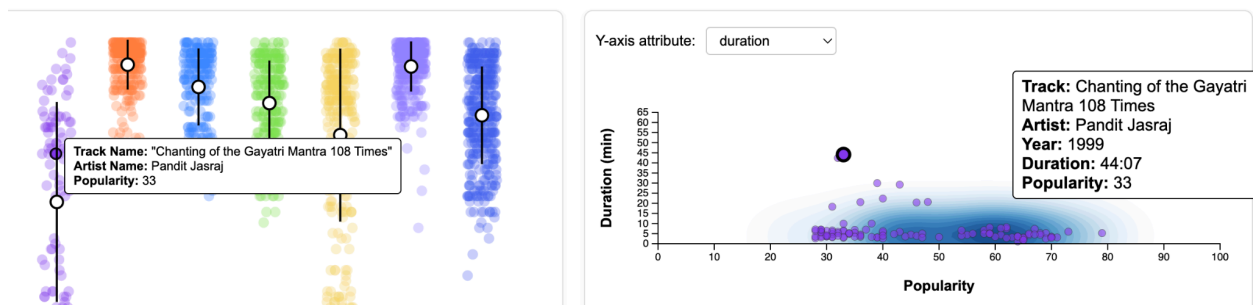
To continue separating the trendlines will implement a mechanic that isolates the feature being hovered over and by obscuring the other two. The obscuring is done by decreasing saturation and increasing brightness, essentially decreasing opacity. To answer our later question about genres, we implemented a genre filter function that only pushes data for a specific genre into the graph.

Implementation

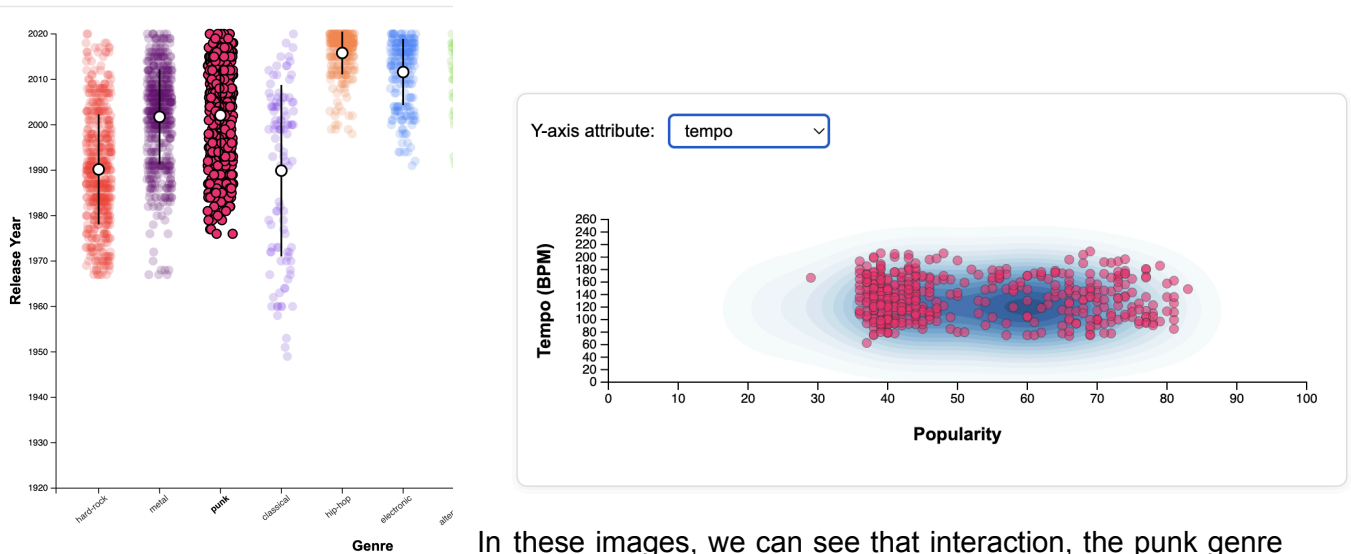
Starting with the largest graph, we have implemented several different interactive features. The purpose of this graph is to highlight how different genres of music have changed in popularity over a decade. So, when hovering over each data point you can see the specific artist, track name, and popularity of each song. Also, if you select a data point on the first graph, it will highlight that track in the second graph.



This image shows the bottom most point being hovered over, although you can't see the cursor due to the screenshot tool.



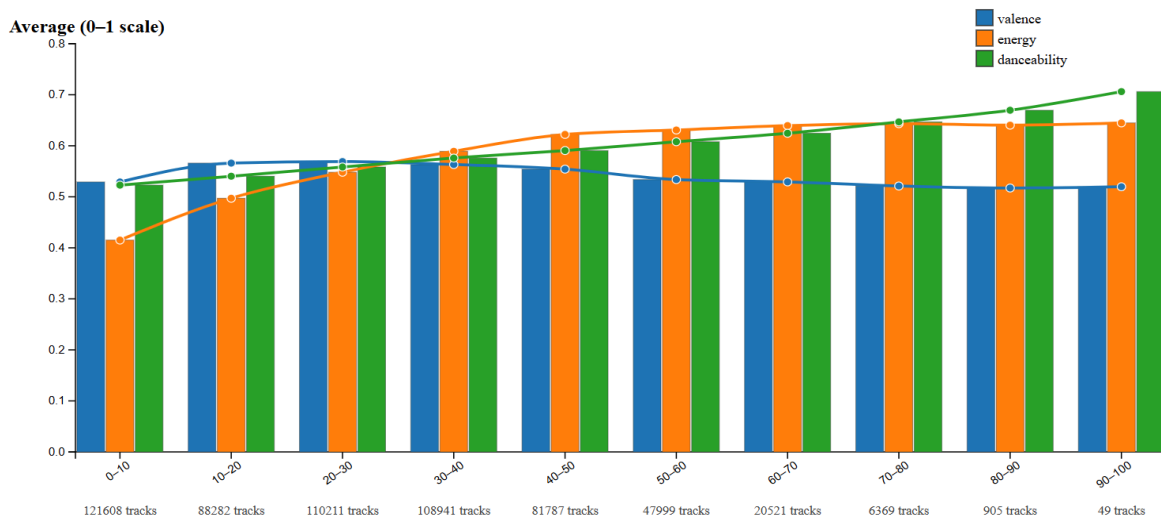
This way you can see how this track compares to the rest of the tracks in its genre in terms of duration, loudness, etc. The second interactive feature of this graph is the genres. By selecting a genre on the x-axis it will highlight all the data points within that genre on the graph. It will also update the other two graphs so they only analyze data points from that specific genre.

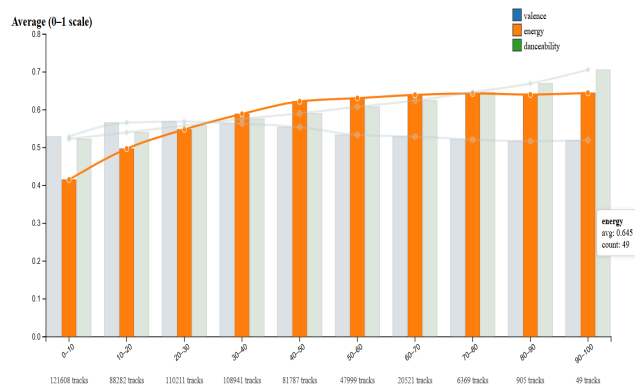


In these images, we can see that interaction, the punk genre has been selected and all those points are shown on the technical features graph, with the chosen y-axis attribute and popularity as their axes.

The second graph interactions are similar to a lot of the interaction implementations in the genres graph. The user is able to hover over the points to see their information, and click on them so that it highlights the point on both this graph and the genre graph. The dropdown menu, as discussed in **Design Evolution**, allows for the user to change the y-axis, to see the different technical attributes and how they might affect the popularity of a song. Overall the implementation and interaction with this graph is simple and similar to the genre graph interactions.

For the third visualization (bottom right), we describe the relationship between popularity and our perceptive features (energy, valence, and dancability). Ten buckets hold songs of popularity scores by the tenth (scores from 0-0.1, 0.1-0.2, etc.). Each bucket has a bar for each feature that is the average value for that bucket. It should be noted that not all buckets have the same number of songs.





This graph features an interaction that visually isolates the features by highlighting the feature that the user is hovering over and obscuring the other two. Overall, the graph allows the viewer to see all three present features and how they change with popularity and interact with each other. With the interaction, the viewer can focus on one feature at a time. Lastly, trend lines on the tops of bars for each feature further emphasizes the trend of individual features.

Selecting a genre from the first graph adds a genre filter to the data in this third graph. This lets the user view trends for these variables by genre.

Our last point of implementation we added was the time slider. This feature adds another aspect of exploration for the user. It affects all the graphs, and allows for them to see the frequency of use for attributes change throughout the years, further allowing for the user to see how that affects the popularity of certain genres and songs. The time slider helps the user answer our question of “How have the musical characteristics of popular tracks changed over the past century?” for themselves through experimentation on our dashboard.

Evaluation

From our graphs a lot of general questions can be answered. For example, graph number one shows how genres have changed over time. We can take classical music as an example. From this graph we can see that the first tracks were released in the 40's and 50's. However, as time continues we see a steady release of tracks up until the 2010's. This shows us that while many of us don't listen to classical music, it has still remained popular enough over the last decade. If you highlight pop music on the graph, you can see that the first few releases were in the 80's, but it didn't start to become super popular until the 90's to the 2010's. While many of the answers can be generally assumed by most people, the graphs show that our assumptions were correct. There is a flaw with this graph however. While we assume the number of releases in a given year means the genre is popular, that can't be known for certain. If we could have found a dataset showing the number of streams this track has received along with its release year, we could better answer the question of whether this genre was truly 'popular' during the period of its release.

Although analyzing different genres' popularity over time is part of our main question, another part is how have musical characteristics of different tracks changed over time. These questions can be answered in our next two graphs.

In the second graph we learn a lot about the different technical features of music and how they affect popularity, especially in contrast to how they affect the music in our dataset overall. For

example, if you look at the tempo of pop music and its popularity, it tends to be more popular than the tempo of the average overall music in the dataset. This can be said for all the different attributes with respect to all the different genres. This graph teaches us more about the specific genres as it shows a more detailed graph for them, but since it allows for the user to view the contour graph underneath, they're able to draw many different types of conclusions.

Using the third graph we found that, moving from least to most popular, energy increases, danceability remains constant, and valance increases. These changes, however, are only marginal for the data overall. Using the genre filter, we saw that overall trends for these features are entirely unrelated to the trend for individual genres. For example, energy decreases among popular songs in the classical genre but increases for popular songs in the pop category. This suggest that the audience for classical music desires less energetic music while pop listeners are looking for more energetic music.

In general, the severe disconnect between the overall trends against popularity and the genre specific trends, suggest that the perceptive characteristics of popular tracks is genre specific. That is to say, listeners want different "vibes" from tracks in different genres.

Any interaction between features that is not explained by the trends of those features is not apparent. For example, popular pop songs seem to have high energy and high valance, however, these two features increase as popularity increases which suggest that they are independently contributing to popularity.

This visualization works very well at displaying the information we have in a way that allows us to answer our questions, however, only in a limited capacity. Having low and variable counts in each bucket introduces a significant issue with statistical precision. In the future I would like to transform the data for this graph by giving the popularity a uniform distribution (which would even the counts buckets) and doing so again for each genre. Another functionality that would increase performance would be to allow the user to change the number of buckets. This would let the user choose how to compromise accuracy and precision.